JIAQI (SCARLETT) HE

Machine Learning · Statistical Modeling · NLP
Python · SQL · PyTorch | Statistics & CS

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EDUCATION

University of Chicago 09/2024 – 12/2026
Bachelor of Science in Statistics GPA: 3.66/4.00
Courses: Machine Learning, Modern Methods in Applied Statistics, Natural Language Processing, Fixed Income

University of Illinois Urbana-Champaign 08/2022 – 05/2024
Bachelor of Science in Statistics & Computer Science GPA: 3.98/4.00
Courses: Statistical Modeling, Time Series Analysis, Stochastic Processes, Probability & Statistics

WORK EXPERIENCE

AI Engineer | **Bank of America** | **UChicago Project Lab** | ***Chicago, IL*** 01/2026 – 03/2026
- Engineered an end-to-end **AI agent** (**LangChain** + **Mistral 14B** via **Ollama**) automating credit-memo QA/QC, cutting review from **14–20 min to 1–3 min** per memo; raised LLM extraction accuracy **57% → 86%** through iterative **prompt engineering**
- Constrained the LLM to extraction only and ran compliance rules in **deterministic Python** for reproducible, auditable verdicts
- Built **OCR** ingestion pipeline parsing nested PDF and embedded images; parallelized image processing for a **30x** speedup

Data Science Intern | **Ant Research — Ant Group (Alibaba)** | ***Shanghai, China*** 08/2025 – 09/2025
- Automated a quarterly pipeline using **DeepSeek** to extract structured preference tags from **12,140+ free-text survey responses**, replacing a manual cycle; segmented **age-cohort patterns** feeding **Alipay's app recommendation model**
- Designed a **Total Factor Productivity** metric from worker salary data, industry-composition weights, and an **AI-adoption** label, delivering leadership its first cross-industry measure of AI's productivity impact

Data Science Research Assistant | **UChicago Data Science Institute** | ***Chicago, IL*** 06/2025 – 08/2025
- Fine-tuned **5 LLMs** for offensive-content detection on a **GPU cluster** using **PyTorch** and **Hugging Face**; achieved **94% accuracy**, outperforming a **GPT-4o** few-shot baseline by **54% F1**
- Led a **6-person team** that designed the annotation schema and cleaned **20,000+ records** in **SQL**/**Python**; reduced false positives **22%** on humor classification via **data augmentation**

Business Intelligence Intern | **Marmon Holdings — A Berkshire Hathaway Company** | ***Skokie, IL*** 06/2024 – 08/2024
- Forecast **110% warehouse overflow** 3 months ahead using **ARIMA** in **Python**, prevented supply chain disruptions
- Automated **POS** processing via multiprocessing, cutting manual workload by **75%**
- Built a **Python ETL pipeline** centralizing outsourced data via **SQL**, enabling cross-unit performance benchmarking
- Designed **5+ interactive Power BI dashboards** tracking revenue and invoices across **3 Berkshire subsidiaries**, giving executives their first cross-unit performance view

PROJECTS

1st Place Agenthon Forecasting Track | **NeurIPS 2026 Poster Presenter** 10/2026
- Ranked **1st of 87 teams** as a solo entrant in **Treasury yield, FX & inflation forecasting**, with a probabilistic model producing Monte Carlo distributions 1–4 days ahead for 3M–30Y yields, G10/EM currencies and CPI releases

4th Place Agenthon Explainability Track 10/2026
- Ranked **4th of 63 teams** with an **LLM financial analysis agent** (NVIDIA Nemotron 120B) that retrieves evidence from SEC filings via **BM25** and predicts earnings growth, credit events and post-earnings moves with a prompt ensemble

Payment Fraud Detection Pipeline | **Python, XGBoost, LightGBM, SHAP, MLflow** 05/2026 – 06/2026
- Built a model that flags fraudulent payments in **590,000 real e-commerce transactions** (3.5% fraud); caught a too-good-to-be-true **0.97 AUC** caused by learning from future data, and retrained on past data only for an honest **0.91**

Support Routing Optimization | **Python, Scikit-learn, XGBoost, Pandas** 03/2024
- Improved customer-support routing precision 73% → **89%** training **Random Forest** and **XGBoost** on **200,000+ call logs**; redesigned routing logic to cut unnecessary escalations **3%**, reducing operational staffing costs

LEADERSHIP

Vice President & Treasurer | **UChicago Table Tennis Club** | ***Chicago, IL*** 05/2025 – Present
- Grew club membership **30% Year-over-Year**; led fundraising generating **$3,000+** for events and NCTTA competitions

SKILLS

ML/Modeling: PyTorch · Scikit-learn · XGBoost · LightGBM · Hugging Face · MLflow · feature engineering · model evaluation (AUC, F1)
NLP/LLM: LLM fine-tuning · LangChain · Prompt Engineering · Claude Code · Cursor
Data/Stats: Python · SQL · Pandas · NumPy · R · ETL · hypothesis testing · time-series (ARIMA) · Power BI · Tableau